

3.18 Q: Kinematics

	Content
Q1	Understand and use the language of kinematics: position; displacement; distance travelled; velocity; speed; acceleration.
	Content
Q2	Understand, use and interpret graphs in kinematics for motion in a straight line: displacement against time and interpretation of gradient; velocity against time and interpretation of gradient and area under the graph.
	Content
Q3	Understand, use and derive the formulae for constant acceleration for motion in a straight line; extend to 2 dimensions using vectors.
	Content
Q4	Use calculus in kinematics for motion in a straight line: $v = \frac{dr}{dt}, a = \frac{dv}{dt} = \frac{d^2r}{dt^2}, r = \int v \, dt, v = \int a \, dt$; extend to 2 dimensions using vectors.
	Content
Q5	Model motion under gravity in a vertical plane using vectors; projectiles.